

1. Data

2. Aggregate

3. Embed

4. Align

5. Quantify

Single-cell
atlases
(Human, Mouse)



Cell-type
centroids
(35 types $\times$ 2 spp)



Joint PCA
(33 components,
95.2% variance)



Procrustes
superimposition
(rotation + scaling)



Per-type
residuals
→ divergence ranking

Permutation test: shuffle cell type labels → null distribution → $p < 1e-6$ (1M permutations)